UNIVERSITY OF PERADENIYA

Faculty of Engineering

FINAL EXAMINATION - July 2022

EE 669 - OPTIMIZATION IN COMMUNICATION SYSTEMS

(Four Hours)

Instructions to Candidates:

1. This is an open-book examination.
2. The notation is consistent with that used in lectures.
3. This paper has 4 questions.
4. Answer *all* questions.
5. This paper contains 4 pages.

Question 1 (Convex Sets and Functions) ..... (25pts)

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(1) Is the set $\mathcal{S} = \mathcal{S}_1 \cup \mathcal{S}_2$, where $\mathcal{S}_1 = (-\infty, 5]$ and $\mathcal{S}_2 = [6, \infty)$ convex? Justify your answer. (5pts)

(2) Consider the set $\mathcal{X} \subseteq \mathbb{R}^2$ defined by the following inequalities:

$$[x_1 \geq x_2 \text{ and } x_1 \geq -x_2] \text{ or } [x_1 \leq x_2 \text{ and } x_1 \leq -x_2]\,.$$

   (a) Draw the set $\mathcal{X}$. (5pts)

   (b) Is $\mathcal{X}$ convex? Justify your answer. (5pts)

   (c) What is the convex hull of $\mathcal{X}$? (5pts)

(3) Consider the function $f(x) = ||x - b||_2^2$, where $b \in \mathbb{R}^m$. Show that f is convex by restricting it to a line. Recall that $||\cdot||_2$ denotes that ℓ_2-norm and $||a||_2^2 = a^{\mathrm{T}}a$. (5pts)

Question 2 (KKT Optimality Conditions) ..... (25pts)

Consider the quadratic optimization problem

$$\begin{array}{ll} \text{minimize} & x^{\mathrm{T}} P^{-1} x + q^{\mathrm{T}} x + r \\ \text{subject to} & l \leq x \leq u \end{array} \tag{1}$$

where the decision variable is $x \in \mathbb{R}^2$,

$$P = \begin{bmatrix} 1 & 0 \\ 0 & 9 \end{bmatrix}, \quad q = \begin{bmatrix} -10 \\ -2/3 \end{bmatrix}, \quad r = 26, \quad l = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \text{ and } \quad u = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

(1) Derive the KKT optimality conditions for the problem above. (10pts)

(2) Check whether the primal variable

$$\tilde{x} = (2, 1)$$

and the dual variables

$$\tilde{\lambda} = (0, 0), \quad \tilde{\mu} = (6, 4/9)$$

satisfy the KKT optimality conditions for problem (1), where $\tilde{\lambda}$ is the Lagrange multiplier vector associated with $l - x \leq 0$ and $\tilde{\mu}$ is the Lagrange multiplier vector associated with $x - u \leq 0$. (10pts)

(3) Can $\tilde{x}$ be optimal for the primal problem (1)? Justify your answer. (5pts)

Question 3 (Primal Problem and the Associated Dual Problem)$\cdots\cdots$ (25pts)

Consider the following optimization problem

$$\begin{array}{ll} \text{minimize} & \frac{1}{2}\, x^{\mathrm{T}} R x + s^{\mathrm{T}} x \\ \text{subject to} & Cx \preceq d \end{array} \tag{2}$$

with variable $x \in \mathbb{R}^n$. The problem parameters are $R \in S^n_{++}$, $C \in \mathbb{R}^{m\times n}$, $s \in \mathbb{R}^n$, and $d \in \mathbb{R}^m$.

(1) Let λ denote the Lagrange multiplier vector associated with the constraint. What is the Lagrangian $L(x, \lambda)$ associated with the problem (2)? [5pts]

(2) Determine the Lagrangian dual function g. [5pts]

(3) What is the associated dual problem? [5pts]

(4) Suppose the problem data are as follows: $R = I$,

$$s = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \quad d = \begin{bmatrix} -1 \\ -2 \end{bmatrix}.$$

In this respect, determine $\lambda^\star$, $d^\star$, $x^\star$, and $p^\star$, the dual solution, the optimal value of the dual problem, the primal solution, and the optimal value of the primal problem, respectively. [10pts]

QUESTION 4 (OPTIMAL JAMMING POWER ALLOCATION)····· (25pts)

A set of n jammers transmit with powers $p_1, \ldots, p_n$, where p_i is the power of i-th jammer. Of course, the transmit powers are nonnegative. In addition, the transmit power of each jammer is limited by the maximum transmit power $p_{\max}$. The jammers produce interference power at m receivers, given by

$$d_i = \sum_{j=1}^{n} G_{ij} p_j, \quad i = 1, \ldots, m$$

where G_{ij} is the nonnegative channel gain from jammer j to receiver i. The channel gains G_{ij} for all i, j are known to the jammers. You may assume that the receiver i has a rate R_i (in bits/s) given by

$$R_i(d_i) = \log\left(1 + \frac{\beta_i}{\sigma^2 + d_i}\right), \quad i = 1, \ldots, m$$

where β_i is the received signal power and σ^2 is the noise power at receiver i.

(1) Show that R_i is convex in d_i. (5pts)

(2) Explain how to choose $p = (p_1, \ldots, p_n)$ to minimize the sum rate R given by

$$R(d_1, \ldots, d_m) = \sum_{i=1}^{m} R_i(d_i).$$

using *convex* optimization. You may assume that the parameters β_i and σ^2 of R_i for all i are known by the jammers and jammers can perform the optimization jointly. The formulation of the optimization problem, together with the objective function and the associated constraint functions are to be explicitly mentioned. (20pts)
Hint: The decision variables of the above problem is p and $(d_1, \ldots, d_m)$.

END OF THE QUESTION PAPER